

Lab-9 Ex4 - Variance

A file of numbers is given as **numbers.txt**. There are total of 100 integers separated by newlines.

In this practice, you are going to calculate (population) variance of a **sublist** of numbers without using array. The key idea is $\text{Var}[X] = E[X^2] - E[X]^2$. The user will input *i* and *j* to indicate the start and end of the sublist.

- Open the file, **numbers.txt**, using read mode.
- Initialize `sum_of_square = 0`, `sum = 0`, `N = 0`.
- Repeat the following steps to read the *i*-th number until the *j*-th number is read:
 - Read a number from the file. If it is not the [*i*, *j*]-th number, then continue.
 - Add the number to `sum`.
 - Add the `number × number` to `sum_of_square`.
 - Increase `N` by 1
- Calculate `variance = sum_of_square / N - (sum / N)^2`.
- Output variance up to 4 decimal places.
- Close the file.

You can assume *i* and *j* are within 1 to 100, and $i \leq j$. User-input numbers are ***bold and italic*** in the following sample runs.

Sample run #1 (Variance of the first & second number in the file: 39, 79)

1 2

400.0000

Sample run #2

10 15

183.2222

Sample run #3

1 1

0.0000